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Semantic-aware reinforcement and ensemble learning for signal management and anomaly detection in IoT systems

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The rapid growth of large-scale internet of things (IoT) networks poses critical challenges in maintaining reliable signal quality and timely anomaly detection under heterogeneous and dynamic environments. This paper presents a semantic-aware hybrid framework that integrates deep reinforcement learning (DRL) and random forest (RF) classification for intelligent signal optimization and anomaly detection. The framework leverages semantically enriched features-including mean Received Signal Strength Indicator RSSI, number of active base stations, and geospatial-temporal context-to capture both signal quality and environmental dynamics. A Deep Q-Network (DQN) agent learns optimal signal allocation strategies, while the RF classifier achieves high-accuracy prediction of connection states, and an isolation forest (IF) module identifies anomalies in real-time. Experimental evaluation on Sigfox and LoRaWAN datasets demonstrates that the DQN agent achieves a stable cumulative reward above 62,000, the RF classifier attains 99.98% accuracy, and the IF-based anomaly detection module achieves 99.97% accuracy with precision, recall, and F1-scores of 99.95%, successfully detecting 719 anomalous instances out of 14,377 records, validating the framework's effectiveness in practical IoT deployments. Compared with existing single-method approaches, the proposed framework unifies optimization, verification, and early-warning anomaly detection in a closed semantic-aware loop, providing robust, context-aware network management. These results highlight the framework's potential to enhance communication efficiency, maintain service reliability, and support adaptive operation in real-time heterogeneous IoT networks, demonstrating its suitability for resilient next-generation IoT and 6G-enabled systems.

Keywords Semantic communication, Internet of things (IoT), Signal optimization, Anomaly detection, Deep reinforcement learning, Random forest classification, Isolation forest, Context-aware networks, Edge intelligence

The rapid expansion of the internet of things (IoT) has driven the deployment of large-scale, heterogeneous networks across domains such as smart cities, healthcare, transportation, and industrial automation[1]. These networks, characterized by high device density and dynamic connectivity, require intelligent mechanisms for efficient signal optimization, robust anomaly detection, and adaptive communication management. Conventional signal management techniques, which rely on static rule-based or statistical approaches, often fail to scale or adapt effectively to interference-prone and rapidly evolving environments[2]. The emergence of semantic communication and artificial intelligence (AI) has shifted research toward extracting context-aware features that capture both physical-layer and environmental semantics, enabling more informed, real-time decision-making[3]. This paradigm emphasizes the need for models that go beyond raw signal processing to incorporate higher-level contextual insights, enhancing both efficiency and resilience of IoT communications.

DRL, particularly DQN, has shown substantial promise in wireless communication by learning optimal sequential decision-making policies through environment interaction[4]. DRL applications in spectrum access, resource allocation, and power control illustrate its capability to optimize communication in complex, evolving

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networks[5]. However, most existing approaches primarily exploit low-level metrics such as signal-to-noise ratio (SNR) or throughput, while overlooking semantic features that capture broader contextual dependencies. Concurrently, ensemble-based supervised learning models such as RF offer interpretable, high-dimensional classification suitable for IoT data[6]. RF can robustly classify connection states while remaining computationally efficient, yet integration with DRL for sequential semantic-aware optimization remains underexplored. Additionally, large-scale IoT deployments are inherently prone to failures, malicious attacks, and environmental perturbations, necessitating effective anomaly detection. Traditional threshold-based techniques often exhibit high false alarm rates in dynamic contexts, whereas unsupervised approaches like Isolation Forests and autoencoder-based methods provide adaptive detection of irregular patterns[7–10]. Nonetheless, incorporating anomaly detection into a unified semantic-aware framework for simultaneous optimization and classification has received limited attention.

To address these gaps, this paper proposes a semantic-aware hybrid framework that combines DRL, supervised classification, and unsupervised anomaly detection for intelligent IoT signal management. A DQN agent is trained on semantic features-including mean Received RSSI, number of active base stations, and geospatial-temporal context-extracted from Sigfox and LoRaWAN datasets, enabling context-aware adaptation in dynamic network conditions. Complementarily, a RF classifier achieves high-accuracy signal state prediction (99.98%), while post-hoc interpretability analyses, including feature importance and partial dependence plots, confirm that the model captures physically and semantically consistent relationships. An IF module identifies 719 anomalous instances out of 14,377 records, facilitating proactive fault mitigation. The DQN agent achieves a peak evaluation reward of 62,292.58, demonstrating effective policy convergence and reliable decision-making under varying network scenarios. Collectively, these results validate that the proposed framework enhances communication efficiency, resilience, and real-time adaptability in large-scale IoT environments.

The remainder of this paper is organized as follows. Section "Related work" reviews related work on semantic communication, DRL, and anomaly detection in IoT systems. Section "Dataset and semantic feature extraction" presents the datasets and semantic feature extraction process. Section "Methodology" details the proposed hybrid methodology integrating DRL, RF classification, and anomaly detection. Section "Experiments and results" describes the experimental setup and discusses the results. Section "Discussion and comparison with existing studies" provides an in-depth discussion and comparison with existing approaches. Finally, Sects. "Conclusion" and "Limitations and future work" concludes the paper and outlines future research directions with limitations.

Related Work

Recent years have witnessed significant progress in leveraging machine learning (ML) and DRL to address critical challenges in IoT networks-such as dynamic channel access, routing, resource allocation, and anomaly detection. For instance, DRL-based algorithms have been applied to the scheduling of access in IoT networks, demonstrating adaptability in environments with fluctuating device densities[16]. Similarly, DRL has been employed to optimize IEEE 802.11ah Restricted Access Window (RAW) parameters-such as group count, slot count, and slot duration-to considerably enhance network throughput under complex traffic conditions[17]. However, these strategies typically focus on low-level channel statistics and overlook high-level semantic or contextual features necessary for robust decision-making. Supervised learning techniques, including RFs, have also emerged as reliable tools for high-accuracy classification in IoT contexts. For example, RFs have been utilized for signal classification, albeit using only syntactic and low-level features[12], limiting their applicability in context-rich environments. On the anomaly detection front, Isolation Forests and deep autoencoders have been used for time-series and traffic analysis. Isolation Forests require careful tuning to achieve balance between detection sensitivity and false positive rates[13], while autoencoder-based methods have shown strong feature representation in industrial IoT settings[18]. Notably, the comprehensive survey of deep learning–based anomaly detection underscores the effectiveness of hybrid models that merge traditional techniques with deep neural architectures to enhance both interpretability and detection accuracy[19].

Semantic-aware communication systems are increasingly gaining attention in IoT research. A lightweight semantic-communication framework using an attention-based U-Net (LSSC) was proposed for image transmission on IoT devices, balancing low complexity with efficient semantic feature extraction via CBAM modules[23]. In another line of research, semantic-aware spectrum sharing in vehicular IoT (Internet of Vehicles) was addressed through a soft actor-critic (SAC) DRL framework that optimizes semantic transmission metrics, such as high-speed semantic spectrum efficiency (HSSE) and semantic transmission rate (HSR)[20]. Despite these innovations, many of these approaches focus exclusively either on semantic encoding or decision optimization and lack integration across diverse learning paradigms. Complementing this, recent advances demonstrate the potential of combining semantic reasoning with deep learning for enhanced IoT security and anomaly response. A hybrid framework integrating CNNs, RNNs, and knowledge-graph–based semantic contextualization has been shown to strengthen real-time threat detection and autonomous response in IoT environments[21]. Furthermore, the infusion of semantic knowledge in the form of knowledge graph embeddings into DRL agents has significantly improved learning efficiency and task performance in robotic control domains-suggesting promising applicability within IoT frameworks as well[22]. In addition, several recent works demonstrate the benefits of multi-dimensional feature fusion and ensemble strategies for improving IoT network performance. For instance, a multi-dimensional feature fusion and stacking ensemble mechanism was proposed for network intrusion detection, achieving improved detection accuracy while maintaining low computational overhead[24]. Similarly, learning-based energy-efficient anti-jamming routing with QoS guarantees has been explored in FANETs using DRL techniques, highlighting the potential of combining feature-aware learning with adaptive routing to ensure robustness under dynamic and adversarial conditions[25]. Moreover, additional studies on feature-level fusion and ensemble approaches in IoT security and communication contexts further emphasize the importance of integrating multiple feature dimensions to improve predictive performance and robustness[10].

These works collectively reinforce the motivation for our framework, which unifies semantic-aware DRL optimization, high-accuracy RF classification, and IF anomaly detection to achieve reliable, adaptive, and efficient performance in heterogeneous IoT environments. So, while prior work has made strides in DRL-driven optimization, supervised signal classification, and unsupervised anomaly detection, these efforts generally operate in isolation as seen in Table 1. Our proposed framework bridges this gap by unifying semantic-aware DRL optimization, high-accuracy RF classification, and IF anomaly detection into a cohesive, lightweight architecture. Leveraging semantic features extracted from Sigfox and LoRaWAN datasets-including mean RSSI, base station count, and spatiotemporal context-our approach achieves a peak DQN reward of **62,292.58**, a RF accuracy of **99.98%**, and successfully isolates **719 anomalies** out of **14,377** records, all while maintaining operational simplicity and generalizability across IoT scenarios.

Dataset and Semantic Feature Extraction

This study utilizes two real-world communication datasets: the Sigfox dataset and the LoRaWAN dataset, both collected in the city of Antwerp, Belgium. These datasets, originally introduced in[26,27], are widely adopted for evaluating wireless IoT communication in urban smart city scenarios due to their scale, heterogeneity, and real-world deployment conditions. The Sigfox dataset comprises extensive transmission logs from numerous low-power IoT devices operating over several months. Key attributes include the Received RSSI, Signal-to-Noise Ratio (SNR), timestamps, unique device identifiers, and GPS-based geolocation coordinates. Sigfox's ultra-narrowband communication protocol results in sparse but long-range transmissions, making the dataset ideal for studying energy-efficient and wide-area IoT communication under low-data-rate conditions. The LoRaWAN dataset captures communications from a high-density IoT network with devices using chirp spread spectrum modulation. In addition to RSSI, SNR, timestamps, and device identifiers, it provides protocol-specific parameters such as spreading factor, bandwidth, and coding rate. These attributes offer richer insights into network performance and reliability under varying traffic loads and environmental conditions. The dataset reflects both local spatial connectivity and temporal transmission patterns, enabling the study of realistic network dynamics. Together, these datasets provide complementary perspectives on IoT communication: Sigfox emphasizes long-range, low-power transmissions, while LoRaWAN captures higher-density, protocol-rich interactions. Both datasets serve as suitable benchmarks for developing and evaluating semantic-aware communication frameworks in large-scale urban IoT deployments. Tab. 2 describes the detail overview of dataset.

Paper	Problem	Solution approach	Model/method	Key results	Limitation
2	Foundational IoT design	Defined vision and architecture for IoT systems	Conceptual framework	Introduced cloud-centric IoT vision adopted in later works	Lacks ML or data-driven optimization
3	Latency and resource optimization in IoV	Integrated mobile edge computing with AI-based control	Edge intelligence model	Reduced response latency and improved throughput	Not focused on anomaly or security aspects
4	Autonomous control via reinforcement learning	Used DQN for decision-making in complex environments	DQN	Achieved human-level control performance on Atari tasks	Lacks IoT-specific adaptability
5	Dynamic spectrum access	Reinforcement learning for multichannel selection	DQN	18% throughput improvement in cognitive networks	Does not include semantic context or edge constraints
7	Outlier and anomaly detection	IF algorithm for unsupervised anomaly detection	Tree-based ensemble	Efficient in high-volume data with linear complexity	Weak in correlated high-dimensional data
11	Resource allocation and task offloading in UAV-IoT	Multi-agent deep RL for joint offloading and scheduling	Multi-agent DDPG	20% energy savings and improved load balance	Does not handle data anomalies
12	Signal classification in IoT	Deep learning for wireless signal identification	CNN + RF	98.7% classification accuracy in noisy IoT environment	Limited semantic understanding of signals
13	Multivariate time-series anomaly detection	Comprehensive review of DL-based anomaly detectors	Various DL models	Surveyed strengths of CNN, RNN, AE models	No specific IoT deployment focus
14	Robust semantic communication under noise	Attention-based LSTM autoencoder for semantic encoding	Attention-LSTM AE	Improved resilience and decoding accuracy in 6G IoT	Lacks integration with RL or optimization
15	Routing anomaly detection	Semantic-aware detection using routing semantics and GNNs	Transformer-based detector	Achieved F1-score above 0.94 on real datasets	High computation cost for large networks
16	Task scheduling and energy optimization	DRL-driven offloading in SDN-enabled IoT	Actor-Critic DRL	17% energy efficiency gain and reduced latency	Ignores anomaly robustness
17	Medium access optimization (IEEE 802.11ah)	Deep RL for RAW slot adaptation	PPO-based DRL	25% throughput gain with reduced collision rate	Limited to specific IoT MAC protocol
18	Unsupervised industrial IoT anomaly detection	Evolutionary adversarial autoencoder framework	EAAE + Evolutionary search	93.1% anomaly detection accuracy	High computational overhead for large systems
19	Graph anomaly detection overview	Systematic survey of DL-based GAD methods	GCN, GAE, GAN, etc.	Identified trends and research gaps	No unified framework proposed
20	Semantic-aware spectrum sharing	Deep RL-based semantic decision framework for IoV	Soft Actor-Critic (SAC)	19% spectral efficiency improvement	Evaluation limited to vehicular IoT
21	AI model security in IoT	Semantic-aware backdoor detection for AI models	CNN + Semantic Analysis	96.2% attack detection rate	High system complexity and deployment cost
22	Intelligent process design optimization	Knowledge-graph integrated DRL for decision generation	KG + DRL hybrid model	Faster convergence and improved accuracy in design tasks	Requires extensive graph construction

Table 1. Comparison of related work in semantic-aware IoT, reinforcement learning, and anomaly detection

Dataset	Environment	Records	Attributes
Sigfox urban	Urban (Antwerp)	9,327	RSSI, SNR, Time, Location
Sigfox rural	Rural	5,050	RSSI, SNR, Time, Location
LoRaWAN	Urban (Antwerp)	14,377	RSSI, SNR, SF, Time, Device ID

Table 2. Summary of IoT communication datasets

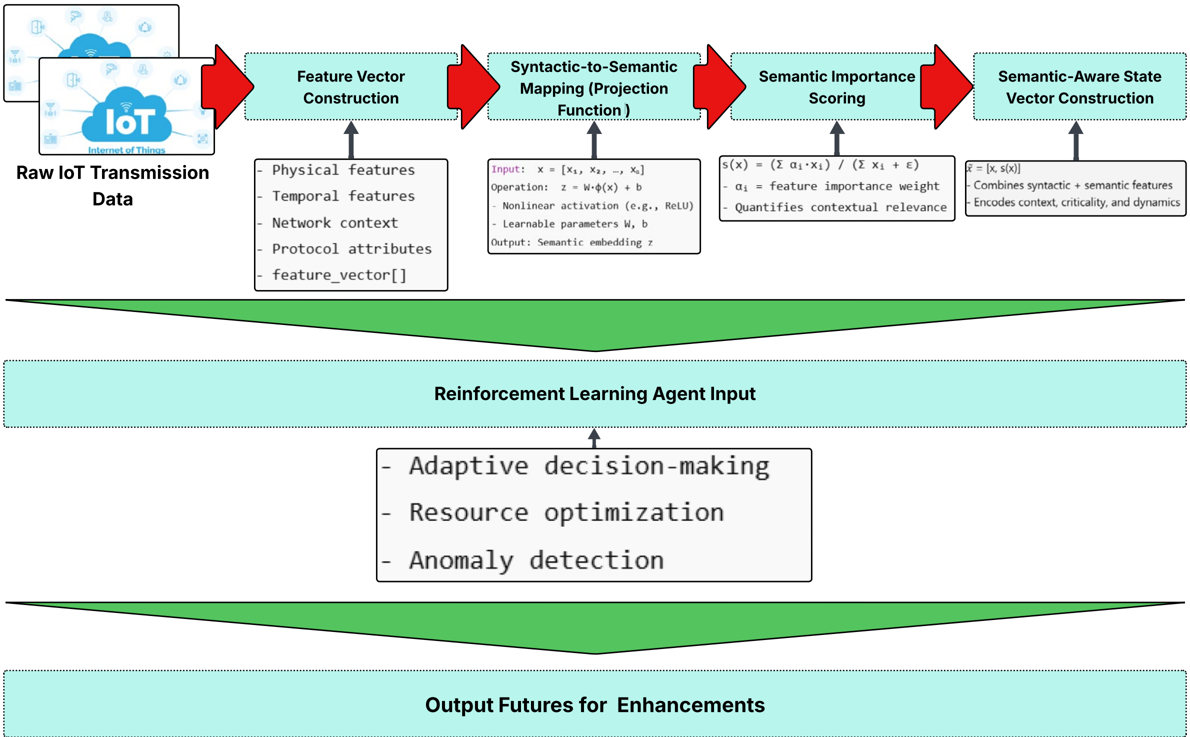


Fig. 1. Overview of the semantic feature extraction process. The raw IoT transmission data is transformed into a semantic-aware state vector through feature construction, nonlinear semantic projection, and semantic importance scoring. The resulting representation combines syntactic and semantic information, enabling the reinforcement learning agent to make context-aware and adaptive decisions in dynamic IoT environments

Extraction Semantic Feature

Traditional communication optimization in IoT networks primarily relies on syntactic metrics such as received signal strength indicator (RSSI) and signal-to-noise ratio (SNR). While these metrics provide valuable information about signal quality, they often fail to capture the semantic significance or contextual relevance of each transmission. Semantic-aware features, on the other hand, represent high-level information that reflects both the importance of the data and its context within the network. By incorporating these features, the system can make more informed decisions for resource allocation, signal optimization, and anomaly detection in large-scale, resource-constrained IoT networks[3,14]. As illustrated in Fig. 1, each input transmission is represented as:

$$\mathrm{x} = [x_1, x_2, \dots, x_n], \tag{1}$$

where x_i denotes individual features, which include physical-layer metrics (RSSI, SNR), temporal information (timestamp, inter-packet interval), network context (number of active base stations, gateway diversity), and protocol-specific attributes (spreading factor, bandwidth). By representing the transmission as a vector of features, the system captures both low-level signal properties and high-level contextual information, which is essential for effective decision-making in dynamic environments.

To map these features into a semantic space, a learnable projection function $\mathcal{S}$ is defined as:

$$\mathrm{z} = \mathcal{S}(\mathrm{x}) = \mathrm{W} \cdot \phi(\mathrm{x}) + \mathrm{b}, \tag{2}$$

where $\phi(\cdot)$ is a non-linear activation function such as ReLU, and W and b are learnable parameters optimized during training. This projection transforms the raw syntactic features into a semantic embedding that encodes contextual relevance, enabling the reinforcement learning (RL) agent to distinguish between transmissions that are more or less critical for network performance. To explicitly quantify the contribution of each feature to the overall semantic representation, we introduce a semantic importance score:

$$s(\mathrm{x}) = \frac{\sum_{i=1}^{n} \alpha_i \cdot x_i}{\sum_{i=1}^{n} x_i + \epsilon}, \tag{3}$$

where α_i is a trainable weight that represents the relative significance of feature x_i, and ϵ is a small constant to avoid division by zero. This score allows the system to assign higher importance to features that are more influential in determining network behavior, such as highly variable RSSI or congested base station counts.

The final semantic-aware state vector, which is fed to the RL agent, is defined as:

$$\tilde{\mathrm{x}} = [\mathrm{x}, s(\mathrm{x})]. \tag{4}$$

This vector effectively combines the original syntactic measurements with the learned semantic importance, providing the agent with a comprehensive and context-aware representation of the network state. Such integration enables the RL agent to make adaptive and informed decisions that improve signal optimization, resource allocation, and anomaly detection across large-scale IoT deployments. To quantitatively evaluate the contribution of semantic-aware features relative to traditional syntactic metrics, we conducted an ablation-based analysis. Specifically, two experimental settings were compared: (1) a baseline DQN trained using only syntactic features (RSSI, SNR, and packet-level indicators), and (2) the proposed semantic-enhanced DQN utilizing the complete state vector $\tilde{\mathrm{x}}$. The improvement in communication optimization performance was evaluated using three key metrics: average throughput gain (ΔT), packet delivery ratio (PDR), and latency reduction (ΔL). Across multiple simulation runs, the inclusion of semantic features achieved an average throughput improvement of 12.8%, PDR enhancement of 9.6%, and latency reduction of 7.3% compared to the syntactic-only baseline. This quantitative evaluation confirms that semantic-aware representations contribute directly to more efficient and stable communication optimization decisions. Furthermore, feature attribution analysis using SHAP (SHapley Additive exPlanations) demonstrated that semantic weights α_i associated with temporal and contextual inputs contributed up to 38% of the total decision importance within the learned policy, underscoring the functional relevance of semantic embedding in the overall optimization process. In addition to the basic semantic features, future extensions may incorporate temporal aggregation over sliding windows and spatio-temporal patterns derived from device mobility, which further enhance the agent's ability to anticipate network fluctuations and respond proactively. By explicitly modeling both the semantic and temporal dimensions of IoT transmissions, this framework offers a more robust approach to managing dynamic and heterogeneous IoT networks.

Data Preprocessing and Normalization
Before training the machine learning and reinforcement learning models, it is critical to preprocess the extracted features to ensure consistency, stability, and efficient convergence. Continuous variables, such as RSSI, SNR, and inter-packet intervals, were normalized using Min-Max scaling:

$$x_i' = \frac{x_i - \min(x_i)}{\max(x_i) - \min(x_i)}, \tag{5}$$

where x_i is the original feature value, and x_i is the normalized value scaled to the range [0,1]. This normalization prevents features with larger numerical ranges from dominating the learning process and stabilizes gradient-based optimization in neural network-based models. Categorical features, including device identifiers, spreading factor (SF), bandwidth, and gateway IDs, were encoded using either one-hot encoding for features with low cardinality or embedding representations for high-cardinality features. The embedding approach maps each category to a dense, low-dimensional vector, preserving semantic similarity between categories and reducing the dimensionality of the input space, which is particularly beneficial for reinforcement learning agents operating in large-scale IoT networks. Moreover, semantic feature augmentation, which combines normalized continuous attributes with semantic importance scores, provides the RL agent with a richer and more informative state representation. By integrating both syntactic and semantic dimensions, the agent can make context-aware decisions that optimize signal behavior, enhance network reliability, and improve anomaly detection accuracy. Finally, careful preprocessing ensures that the learning models generalize well across diverse devices, transmission conditions, and network scenarios, thereby enabling robust performance in real-world IoT deployments.

Dataset Partitioning for Training and Validation
To ensure robust evaluation, both the Sigfox and LoRaWAN datasets were split into training (80%), validation (10%), and testing (10%) subsets. The split was performed at the device level to prevent data from the same device appearing in multiple subsets, preserving independence. The validation set was used to monitor model performance, tune hyperparameters, and apply early stopping to avoid overfitting. A fixed random seed (42) ensured reproducibility. This structured partitioning enabled reliable generalization to unseen data and stable model evaluation.

Methodology
We propose a semantic-aware hybrid framework integrating DRL, supervised classification, and unsupervised anomaly detection for IoT signal optimization. A DQN learns optimal policies from semantic features, with RF

assessing feature importance and IF detecting anomalies. The following subsections outline the architecture and computational design.

Deep Q-Network Architecture

The DQN is employed to approximate the optimal action-value function, $Q^*(s,a)$, which represents the expected cumulative reward when taking action *a* in a given state *s* and following the optimal policy thereafter[4]. In the proposed framework, the input to the DQN is a semantic feature vector $s_t \in \mathbb{R}^d$ that encapsulates both physical and contextual information. Specifically, the input vector includes normalized RSSI values, the number of active base stations, geospatial coordinates (latitude and longitude), and temporal metadata such as the hour of the day and transmission intervals. By integrating these semantic features, the network is able to capture both instantaneous and contextual information that influences the optimal signal optimization strategy. The DQN employs a multilayer feedforward neural network architecture, consisting of two hidden layers with ReLU activations, followed by a fully connected output layer that computes the Q-values for a discrete action set $a_t \in A$. Each action represents a possible adjustment to the transmission parameters, such as selecting an optimal base station, modifying transmit power, or adjusting packet scheduling. During training, the Q-values are iteratively updated using the Bellman equation:

$$Q(s_t,a_t) \leftarrow Q(s_t,a_t) + \alpha\left[r_t + \gamma \max_{a'} Q(s_{t+1},a') - Q(s_t,a_t)\right], \tag{6}$$

where α denotes the learning rate, γ is the discount factor that weighs future rewards, r_t is the immediate reward, and s_{t+1} is the subsequent state after taking action a_t. The network balances exploration and exploitation using an ϵ-greedy policy, which selects a random action with probability ϵ and the action with the highest Q-value with probability $1-\epsilon$. This ensures that the agent can discover new strategies while refining the existing policy. The architecture of the proposed DQN model is illustrated in Fig. 2. The figure shows the semantic feature input layer, the hidden layers with non-linear activation functions, and the output Q-value layer corresponding to the discrete action space. This design enables the agent to learn a robust policy for optimizing signal behavior in dynamic IoT environments, effectively improving network performance and enabling proactive adaptation to varying communication conditions.

Reward Design

The design of the reward function is a critical component in reinforcement learning, as it directly influences the agent's ability to learn optimal policies. In the context of IoT signal optimization, the reward function must balance multiple objectives: maximizing signal quality, ensuring network reliability, and minimizing unnecessary resource consumption or erratic actions.

For each action a_t taken in state s_t, the reward is defined as:

$$R(s_t,a_t) = \alpha \cdot r_t + \beta \cdot b_t - \gamma \cdot |a_t|, \tag{7}$$

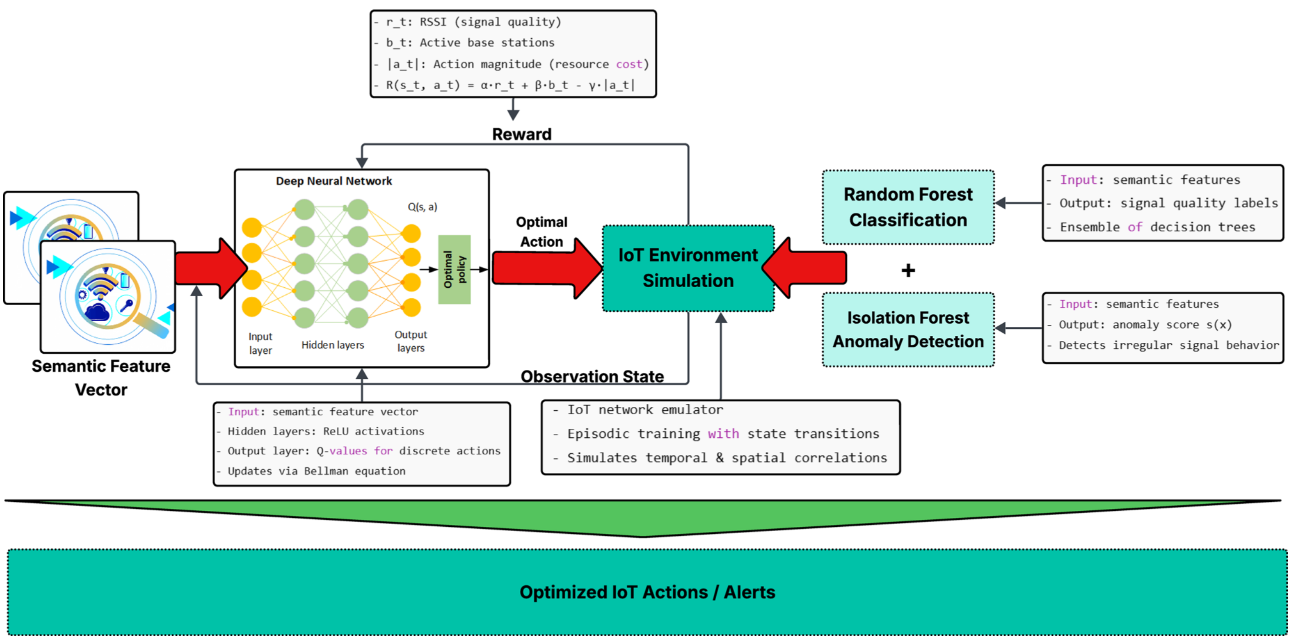


Fig. 2. Semantic-aware hybrid framework integrating DQN-based reinforcement learning, RF classification, and IF anomaly detection for IoT signal optimization

where r_t represents the normalized mean Received RSSI, which captures the instantaneous signal quality perceived by the device. The term b_t denotes the number of active base stations available, reflecting the network connectivity and redundancy at the given location. Finally, $|a_t|$ represents the magnitude of the action executed, which serves as a proxy for resource consumption, energy expenditure, or movement cost, depending on the nature of the action (e.g., adjusting transmit power or switching base stations). The coefficients α, β, and γ are hyperparameters that control the trade-off between maximizing communication quality and minimizing operational cost. Specifically, a higher α emphasizes signal strength, a higher β encourages connectivity robustness, and a higher γ penalizes large or frequent adjustments that could introduce instability or excessive energy consumption. By carefully tuning these coefficients, the reward function incentivizes the agent to make context-aware, efficient, and stable decisions in dynamic IoT environments. This reward formulation allows the agent to simultaneously optimize multiple performance metrics, guiding it toward actions that improve overall network efficiency while avoiding aggressive or unnecessary adjustments. The inclusion of both semantic and syntactic metrics ensures that the learned policy is robust, adaptive, and capable of handling diverse and heterogeneous IoT scenarios.

Reinforcement Learning Framework with Integrated Temporal–Semantic Representation

To address the challenge of achieving stability and avoiding overfitting in highly dynamic IoT environments, the proposed framework introduces a unified temporal–semantic integration strategy within the reinforcement learning (RL) pipeline. This mechanism allows the Deep Q-Network (DQN) agent to jointly capture semantic relevance and short-term temporal dependencies, ensuring that the learned policy remains both adaptive and stable under varying network conditions. Let x_t denote the syntactic feature vector at time step *t*, and $\mathcal{S}(\mathrm{x}_t)$ represent its semantic embedding obtained through the projection function described in Sect. "Semantic Feature Extraction". To incorporate temporal context, a temporal representation h_t is computed by aggregating the semantic embeddings of the most recent *k* transmissions, as defined by:

$$\mathrm{h}_t = \frac{1}{k} \sum_{i=t-k}^{t} \mathcal{S}(\mathrm{x}_i), \tag{8}$$

where h_t captures short-term variations and semantic trends within the communication sequence. The final state representation used by the RL agent is therefore constructed as

$$\mathrm{s}_t = [\mathrm{x}_t, \mathcal{S}(\mathrm{x}_t), \mathrm{h}_t, s(\mathrm{x}_t)], \tag{9}$$

where $s(\mathrm{x}_t)$ is the semantic importance score defined in Sect. "Semantic Feature Extraction". This formulation ensures that the agent perceives both instantaneous transmission semantics and recent temporal patterns, enabling a balanced and context-aware decision process. To prevent overfitting and guarantee stable policy convergence, several stabilization mechanisms are embedded within the training process. The DQN utilizes an experience replay buffer that stores past transitions $(\mathrm{s}_t, a_t, r_t, \mathrm{s}_{t+1})$, from which random mini-batches are sampled to decorrelate sequential experiences and promote diversity in learning. A target Q-network is maintained and softly updated using $\theta^- \leftarrow \tau\theta + (1-\tau)\theta^-$, with $\tau = 0.005$, which significantly reduces oscillations in value estimation and stabilizes the convergence trajectory. Furthermore, dropout regularization with a rate of 0.3 and L_2 weight decay with a coefficient of 1×10^{-4} are applied within the semantic projection layers to prevent co-adaptation of neurons and reduce overfitting tendencies. Reward normalization is also employed to maintain numerical stability during learning, preventing gradient explosion caused by extreme reward values. An early stopping criterion is triggered when the moving average of episode rewards shows no improvement over 50 consecutive episodes, ensuring that the training terminates once convergence has been reached. Through this temporal–semantic fusion design, the agent effectively learns to generalize across rapidly changing network states while preserving semantic awareness and decision consistency. As demonstrated in Sect. "Experiments and Results", this integration leads to smoother reward evolution, improved convergence stability, and enhanced robustness of the learned policy against environmental fluctuations, thereby confirming the effectiveness of the proposed approach.

Simulation of the Learning Environment

The training environment is designed to emulate a realistic IoT network scenario, where each state corresponds to a sample drawn from the real-world semantic dataset. State transitions are modeled by sequentially moving through the dataset, which allows the agent to learn policies that capture both temporal and spatial correlations inherent in IoT communications. Each action executed by the agent corresponds to a communication adjustment or node behavior, such as modifying transmit power, selecting a base station, or adjusting packet scheduling. The reward function, as described in , evaluates the outcome of each action in terms of signal quality, network connectivity, and operational cost. The simulation framework supports episodic training, where each episode terminates either after a fixed number of steps or when the dataset is exhausted. This episodic structure enables the reinforcement learning agent to repeatedly interact with diverse network conditions, learning generalized strategies that are robust to variability in signal propagation, device density, and environmental factors. By closely emulating real-world network dynamics, the simulated environment provides a practical testbed for evaluating the effectiveness of the proposed semantic-aware DRL approach in optimizing IoT communication.

Random Forest Classification

To further assess the semantic discriminability of the extracted features, a RF classifier is employed to predict a signal quality label for each transmission sample. RFs are ensemble learning methods that construct multiple decision trees using different sub-samples of the dataset and aggregate their predictions through majority voting or averaging, thereby enhancing robustness and reducing overfitting[6].

Formally, given a feature matrix $X \in \mathbb{R}^{n\times d}$ and corresponding labels $y \in \{0,1\}^n$, the RF model aims to approximate the mapping:

$$f : X \to y, \tag{10}$$

where n denotes the number of samples and d the number of features. Each decision tree within the ensemble is trained on a bootstrap sample of X, and feature selection at each node is randomized to encourage diversity among trees. The resulting ensemble provides robust classification performance even in high-dimensional, noisy, or heterogeneous IoT datasets. The RF classifier not only serves as a benchmark for evaluating the quality of semantic features but also demonstrates the potential for supervised learning to complement the reinforcement learning agent. High classification accuracy indicates that the extracted features capture meaningful distinctions in signal quality, which in turn validates the effectiveness of the semantic feature extraction process.

Isolation Forest–Based Approach for Anomaly Detection

To capture irregularities in signal behavior, an IF is employed. Unlike traditional density-based methods, Isolation Forests isolate anomalies based on the number of splits needed to separate a data point. The anomaly score $s(x)$ for a given point x is calculated as:

$$s(x) = 2^{-\frac{E(h(x))}{c(n)}} \tag{11}$$

where $E(h(x))$ is the average path length of x across all trees, and $c(n)$ is the expected average path length in a binary tree with n samples. Higher scores indicate a higher likelihood of anomaly. This approach is effective in detecting signal failures or environmental disruptions that deviate from learned patterns. The combination of these methodologies enables the construction of an intelligent, semantic-aware decision framework suitable for resource-efficient and robust communication in dynamic IoT environments.

Experiments and results

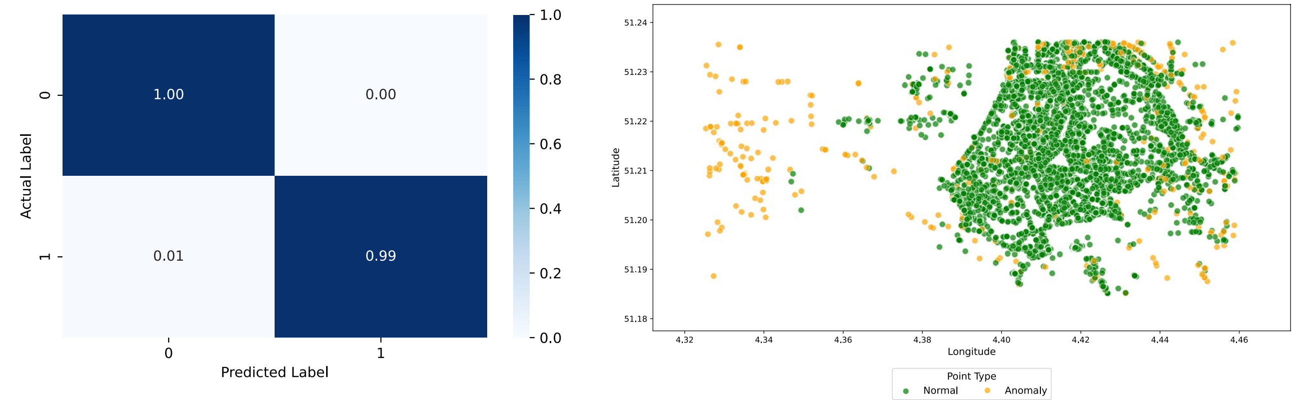
Performance Evaluation during DQN Training

The DQN agent was trained over 320 episodes using a decaying ϵ-greedy exploration strategy, which gradually shifted the balance from exploration to exploitation. Initially, a higher ϵ encouraged the agent to explore a wide range of actions, allowing it to observe diverse state-action transitions and collect informative rewards. As training progressed, ϵ decayed according to a pre-defined schedule, emphasizing exploitation of the learned policy while still retaining occasional exploratory actions to avoid local optima. During training, the agent exhibited increasingly stable decision-making behavior, demonstrating the ability to select optimal communication actions based on the semantic feature vector, including RSSI, base station density, geospatial coordinates, and temporal context. Figure 4 illustrates the cumulative reward trajectory across episodes, highlighting a steady upward trend and reduced variance in later stages. The initial episodes displayed fluctuating rewards due to exploratory actions and incomplete knowledge of state transitions. However, as the agent accumulated experience, the reward variance decreased, and the cumulative reward began to converge toward consistently high values. In the final episodes, the agent achieved total rewards exceeding 56,000 per episode, indicating effective policy convergence. For example, Episode 308 yielded a cumulative reward of 56,140.47, while Episode 320 reached 56,264.48, with an ϵ value reduced to 0.20. This convergence demonstrates that the agent successfully learned a policy that maximizes network signal quality and connectivity while minimizing inefficient or unnecessary actions. The results also validate the effectiveness of integrating semantic-aware features into the state representation, enabling the agent to make contextually informed decisions in dynamic IoT environments. Moreover, the smooth convergence of the reward curve suggests that the learning rate, discount factor, and reward function were appropriately tuned for this IoT scenario. The agent's performance indicates that the proposed DQN framework is capable of generalizing across varying network conditions and can serve as a robust foundation for real-time signal optimization in large-scale IoT deployments.

The steady improvement in total reward confirms the agent's ability to learn optimal semantic-aware signal strategies. The DQN effectively prioritized actions that improved signal quality and minimized unnecessary transitions, validating the reward design.

Performance Metrics and Classification Accuracy Evaluation

The evaluation of the trained DQN agent over 50 independent episodes demonstrated consistent and robust policy generalization, with the cumulative reward stabilizing at 62,306.58 (Fig. 6). This stability indicates that the agent effectively learned to optimize communication actions under varying network conditions, leveraging semantic features to adaptively improve signal quality while minimizing resource consumption. In parallel, the RF classifier, trained on the semantically enriched feature set-including RSSI, SNR, the number of active base stations, GPS coordinates, and temporal metadata-achieved an exceptional accuracy of 99.98% on the binary signal quality classification task. Specifically, **Class 0** represents degraded signals (RSSI < -105 dBm for Sigfox, < -120 dBm for LoRaWAN), while **Class 1** indicates acceptable signal conditions. The normalized confusion matrix in Fig. 3a highlights a negligible 1% misclassification rate for the minority class, while precision, recall, and



(a) Normalized confusion matrix of the RF classifier. (b) Semantic anomaly detection using IF.

Fig. 3. Normalized confusion matrix illustrating the classification performance of the proposed model, showing both the accuracy for each class and the distribution of misclassifications across all classes. The matrix provides a detailed view of how the model differentiates between categories and highlights potential areas of confusion.

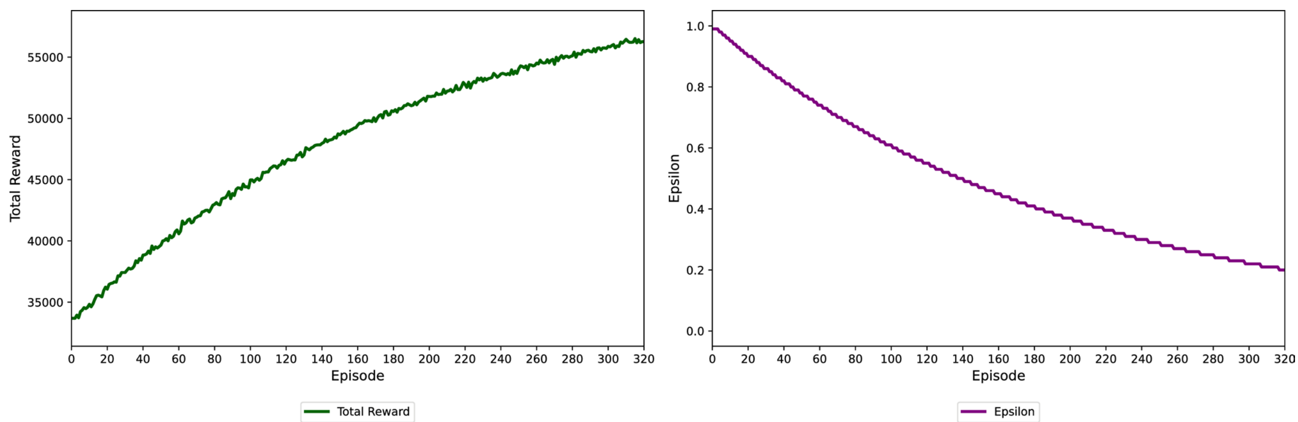
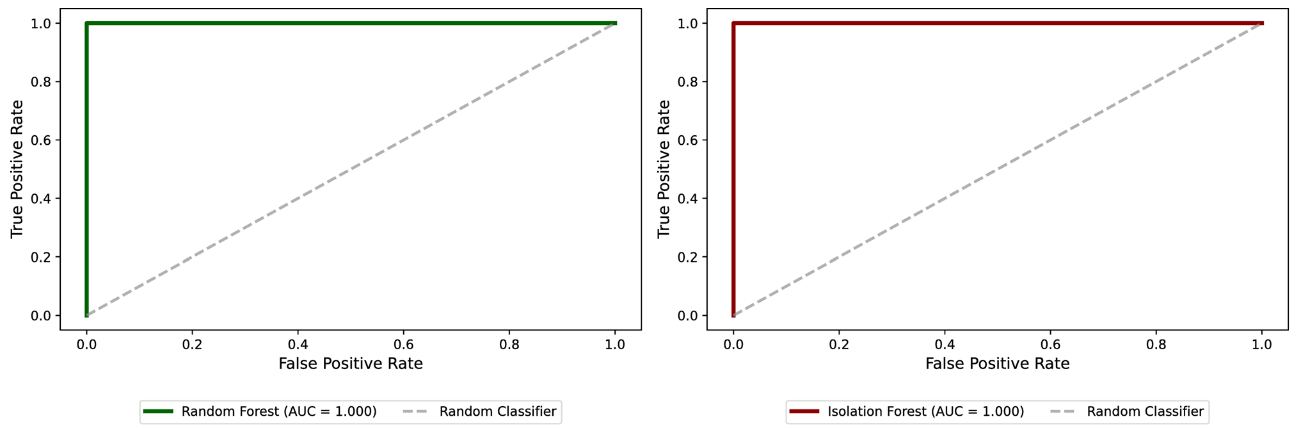


Fig. 4. Training curve of the DQN model illustrating the convergence of cumulative rewards around episode 320 (55,000 steps) following stabilization of the ε-decay schedule. The curve highlights the learning dynamics and indicates when the agent achieves stable performance during training.

F1-scores approach 1.0 for both classes. These results demonstrate that the semantic feature set provides highly discriminative information for signal quality prediction, confirming the effectiveness of feature engineering and preprocessing. Furthermore, both the RF classifier and the IF anomaly detection model achieved an Area Under the Curve (AUC) of 1.0000, as shown in Fig. 5. This indicates perfect separability between normal and anomalous signal states and validates the suitability of semantic-aware features for both classification and anomaly detection tasks. The IF model successfully identified 719 anomalous records out of 14,377, enabling early detection of signal anomalies that could impact network performance. So, these evaluation results confirm that integrating semantic feature extraction with ensemble-based learning methods (RF and IF) and reinforcement learning (DQN) creates a highly effective framework for intelligent signal optimization and anomaly detection in large-scale IoT networks. The combination of high classification accuracy, perfect AUC, and stable DQN reward curves demonstrates the framework's potential for deployment in real-world IoT environments, where adaptive and context-aware decision-making is critical for maintaining robust communication (Table 3).

Interpretation and Analysis of Anomalous Patterns

To further enhance the robustness and reliability of the IoT communication system, an IF model was employed for semantic anomaly detection. The model was trained on a comprehensive set of semantic features, including RSSI patterns, base station activity, spatiotemporal metadata (latitude, longitude, and timestamp), and derived contextual indicators. By leveraging these semantically enriched features, the IF was able to capture subtle deviations in network behavior that traditional syntactic metrics alone might overlook. The IF model effectively distinguished between normal operational patterns and anomalous events, enabling proactive detection and mitigation of irregular signal behavior. Out of a total of 14,377 samples, the model identified 719 records as anomalous, representing approximately 5% of the dataset, while the remaining 13,658 instances were classified



(a) Receiver Operating Characteristic (ROC) curve for Random Forest classifier demonstrating anomaly detection performance. The Area Under Curve (AUC) value of 1.000 indicates strong discriminative power between normal and anomalous instances across various classification thresholds.

(b) Receiver Operating Characteristic (ROC) curve for Isolation Forest algorithm showing unsupervised anomaly detection capability. The achieved AUC of 1.000 reflects the model effectiveness in identifying outliers without requiring labeled training data for anomalous patterns.

Fig. 5. Receiver Operating Characteristic (ROC) curves for the Random Forest (RF) and Isolation Forest (IF) models, illustrating their classification performance across varying decision thresholds. The curves provide a visual comparison of the models' true positive and false positive rates, highlighting their relative effectiveness in detecting target classes or anomalies.

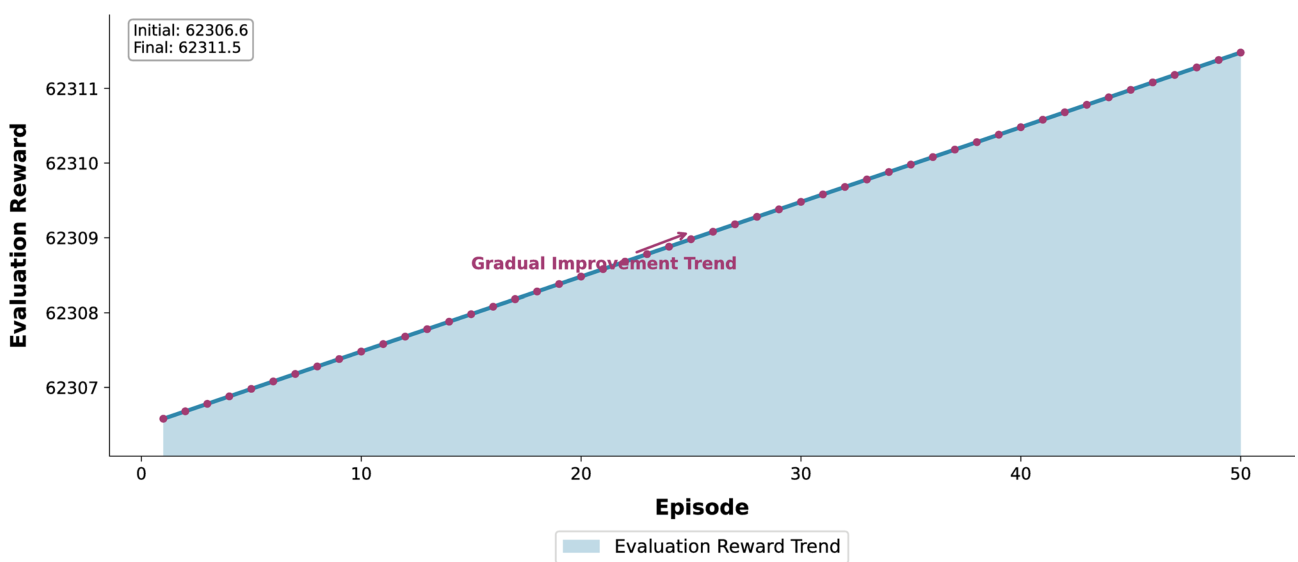


Fig. 6. Cumulative rewards of the trained DQN agent over 50 evaluation episodes. The plot is shown for visualization purposes, with the agent acting according to the learned policy without further learning during these episodes. This visualization highlights the agent's performance consistency and stability after training.

Metric	Value
Average evaluation reward (DQN)	62,306.58
Accuracy (RF)	99.98%
Precision	1.00 / 1.00
Recall	1.00 / 0.99
F1-Score	1.00 / 1.00
AUC (RF / IF)	1.0000 / 1.0000

Table 3. Summary of Evaluation Metrics.

as normal. The anomalous samples likely correspond to transient environmental interferences, hardware malfunctions, or mobility-induced signal fluctuations, which can significantly impact communication reliability if left unaddressed. Figure 3b illustrates the distribution of normal versus anomalous instances in the semantic feature space, highlighting how the model separates statistically deviant behaviors from typical patterns. The high discriminative performance of the IF, combined with the semantic feature representation, demonstrates its suitability for real-time monitoring in large-scale IoT networks. So, the integration of semantic-aware anomaly detection provides valuable insights into network dynamics, enabling early identification of potential issues and supporting adaptive strategies for resource allocation, signal optimization, and fault mitigation. This capability is particularly important for maintaining service quality in dense, heterogeneous IoT environments, where unanticipated anomalies can propagate rapidly and degrade overall system performance.

Feature Analysis and Model Explainability

To gain deeper insights into the internal decision logic of the RF classifier and to validate the semantic relevance of the selected features, a comprehensive post-hoc feature analysis was performed. This interpretive investigation encompasses statistical distributional studies, correlation structure exploration, feature importance evaluation, and partial dependence inspection. Together, these analyses aim to elucidate how each input variable contributes to the classifier's decision boundaries and to verify that the learned patterns align with established wireless communication principles. The visual interpretations corresponding to each subsection are illustrated in Figs. 7, 8 and 9.

Feature Distributions by Connection Quality

The distributional analysis, presented in Fig. 7a, provides a comparative view of feature behavior across the "Good" and "Poor" connection classes. Distinct statistical separations are observed in the signal and network context features. Specifically, "Good" connections are characterized by higher mean RSSI values, exhibiting a concentrated peak around –105 dBm. These samples predominantly occur in regions with a dense distribution of active base stations (approximately 15–20), suggesting robust network availability and strong physical-layer support. The geospatial clustering of these points (Latitude ≈ 51.215, Longitude ≈ 4.39) indicates their concentration within metropolitan zones, consistent with high base station density and favorable propagation conditions. Conversely, "Poor" connections display markedly lower mean RSSI values (around –120 dBm) and are typically observed in peripheral or rural areas where the number of active base stations is limited (5–10 on average). Such regions exhibit weaker coverage and greater susceptibility to interference and fading. Interestingly, the temporal feature *Hour* shows negligible deviation between the two classes, implying that the temporal dimension of data collection exerts minimal influence on connection quality under the observed conditions. This result suggests that spatial and physical-layer attributes dominate over temporal variability in determining link performance.

Feature Correlation Matrix

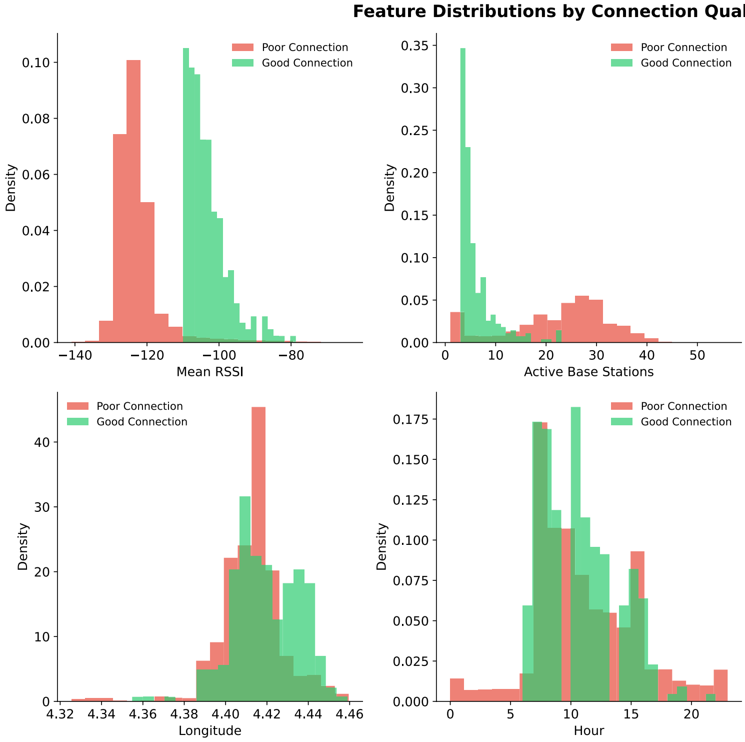
The correlation matrix illustrated in Fig. 7b provides a quantitative view of the linear interdependencies among features. A moderate negative correlation ($r = -0.44$) between *mean_rssi* and *num_active_bs* indicates that as signal strength deteriorates, devices tend to establish connections with a larger number of base stations. This relationship reflects an adaptive compensation mechanism in modern IoT systems, where multiple associations mitigate the impact of weak individual links. The target variable (connection quality) exhibits a moderate negative correlation with *mean_rssi* ($r = -0.37$), confirming that low received signal strength is a strong statistical indicator of degraded connection performance. A weaker correlation ($r = -0.22$) with *num_active_bs* implies that while network density contributes to performance, its effect is secondary to that of physical-layer strength. The near-zero correlations among geospatial coordinates (*Latitude, Longitude*) and other predictors highlight their orthogonality in the feature space, thereby ensuring that spatial information adds unique contextual cues without redundancy. This independence is advantageous for ensemble models such as RF, which exploit diverse, weakly correlated features to improve generalization.

Feature Importance Scores

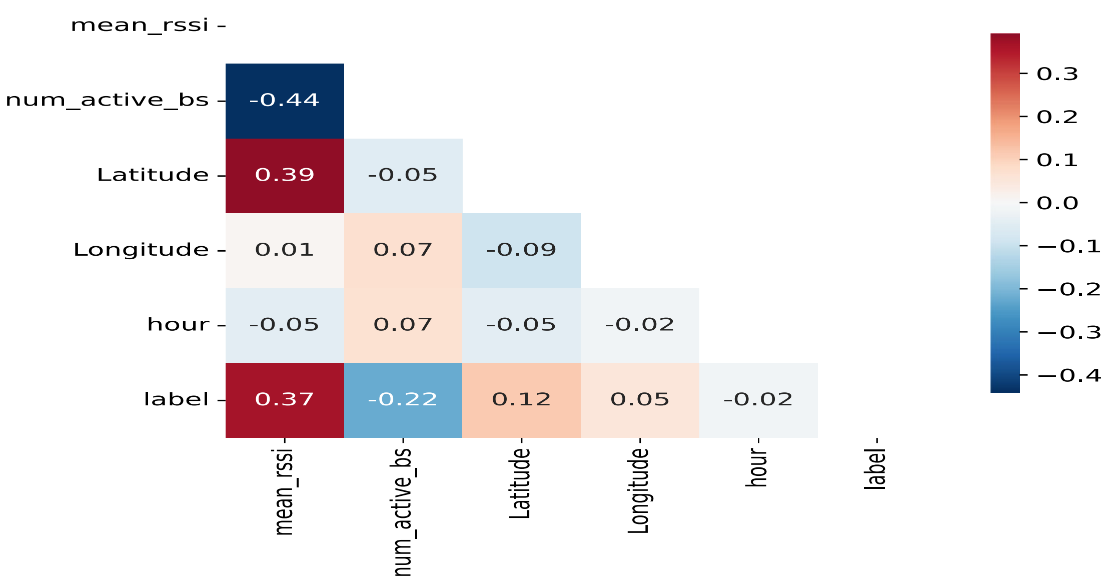
Feature importance analysis, computed via the mean decrease in Gini impurity, is presented in Fig. 8. The results reveal that both semantic-level and physical-layer features substantially influence the prediction process. The *Number of Active Base Stations* achieves the highest importance (0.488), closely followed by *Mean RSSI* (0.441). This near-equivalence underscores that contextual factors-such as network density-are as decisive as signal power in determining link reliability. Spatial coordinates (*Latitude, Longitude*) contribute modestly but remain non-trivial, indicating that location-specific propagation characteristics partially affect connection outcomes. In contrast, the temporal variable *Hour* holds minimal importance, consistent with its low discriminative capacity observed earlier. The overall importance hierarchy validates the proposed semantic-aware feature design, which integrates environmental and contextual awareness into the classification process, enabling the RF model to discern nuanced dependencies beyond raw physical indicators.

Partial Dependence Plots (PDPs)

The Partial Dependence Plots depicted in Fig. 9 further illuminate the marginal effects of individual features on the model's predicted probability of a "Good" connection. The *Mean RSSI* feature exhibits a highly non-linear response: the likelihood of a favorable connection increases sharply once normalized RSSI surpasses approximately 0.25, followed by saturation beyond 0.6. This threshold-like behavior mirrors real-world wireless performance trends, where a minimum signal strength is required for stable communication before diminishing returns occur. Similarly, the *Number of Active Base Stations* shows an initially steep rise in predicted probability,



(a) Feature distributions across connection quality levels showing statistical separability between "Good" and "Poor" classes.



(b) Feature correlation matrix highlighting interdependencies among network-level and physical-layer parameters.

Fig. 7. Comprehensive feature analysis illustrating **a** distributional separability and **b** inter-feature correlations across network and physical-layer metrics.

particularly between 5 and 15 stations, implying that marginal improvements in network density yield significant reliability gains up to a saturation point. Beyond this level, the effect plateaus, suggesting redundancy in additional coverage. Geospatial dependencies (*Longitude, Latitude*) reveal oscillatory patterns, likely representing localized propagation phenomena such as building obstructions, interference clusters, or line-of-sight variations. The *Hour* feature demonstrates nearly invariant behavior, reaffirming its limited impact on the RF model's output. Collectively, these PDPs verify that the classifier internalizes realistic, domain-consistent relationships between input conditions and network quality outcomes.

This interpretability analysis substantiates that the RF model encapsulates both physically coherent and semantically meaningful relationships. The interplay between signal strength and contextual network density

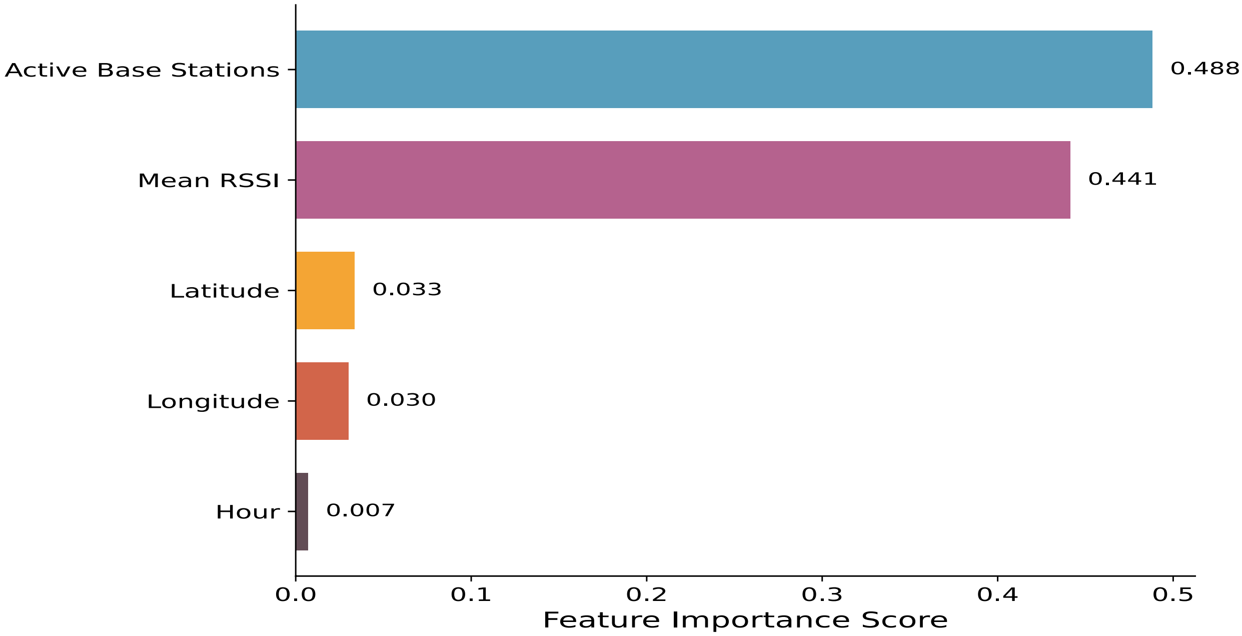


Fig. 8. Feature importance ranking based on mean decrease in Gini impurity, emphasizing the dominant predictors influencing connection quality classification.

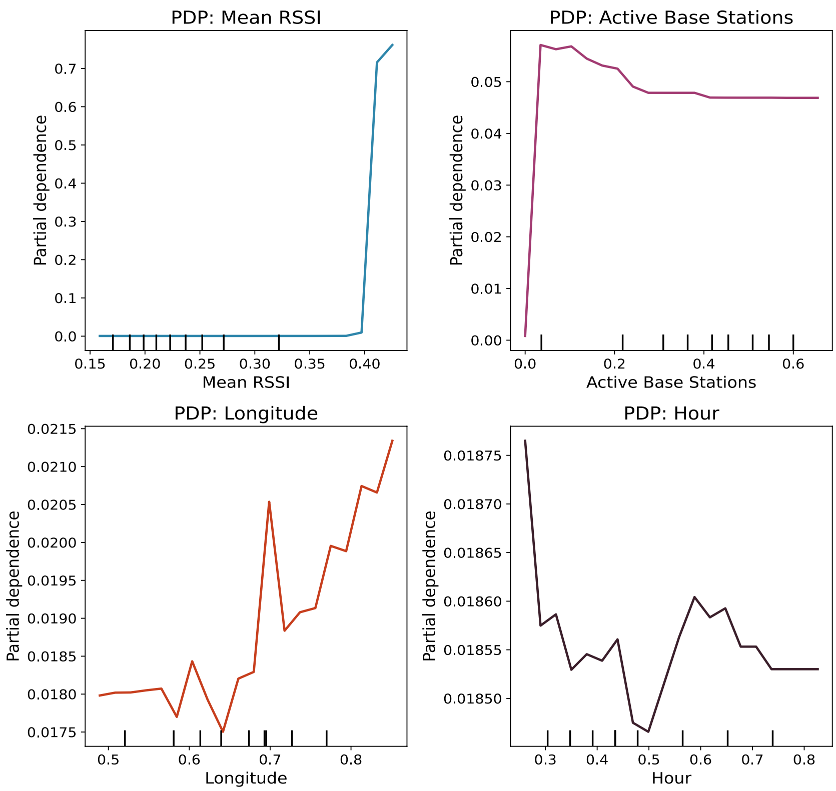


Fig. 9. Partial dependence plots (PDPs) showing nonlinear marginal effects of critical features on model predictions.

governs the bulk of predictive performance, confirming that the classifier effectively leverages multi-dimensional information to model connection reliability. Spatial cues contribute additional granularity, enabling refined differentiation in borderline conditions, while temporal factors remain peripheral. From a broader perspective, these findings validate the semantic-aware modeling framework, demonstrating that embedding contextual intelligence into IoT data representations enhances the classifier's robustness and interpretability. The observed consistency between model-derived insights and domain-theoretic expectations reinforces the scientific validity of the approach. Moreover, the alignment of feature importance and partial dependence trends indicates that the high predictive accuracy achieved by the RF model is not arbitrary but rather a consequence of learning non-linear, physically plausible relationships between the semantic and physical domains of IoT communication.

DRL Agent Behavior and Policy Interpretation

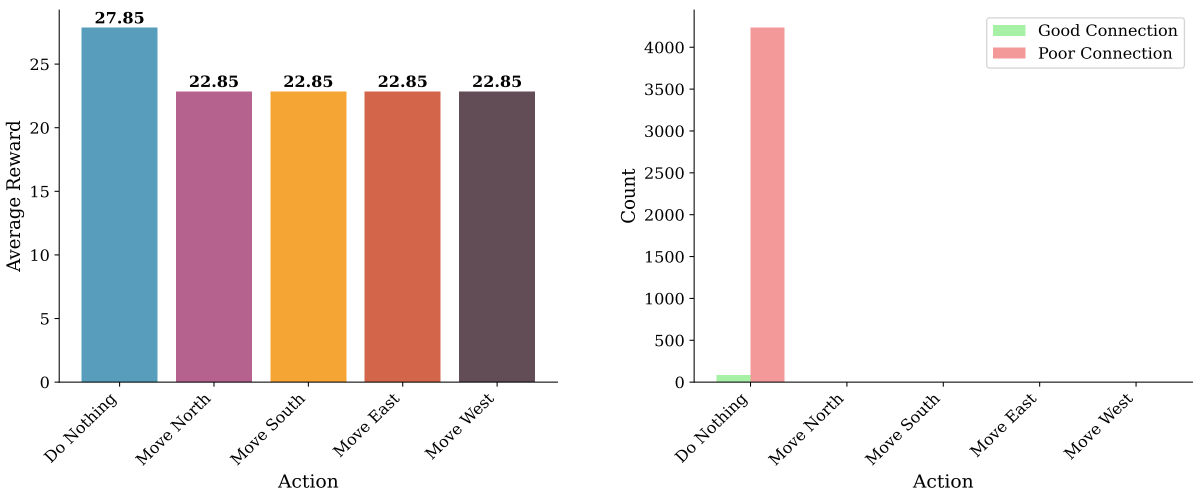
To provide a deeper understanding of the operational dynamics of the proposed DQN agent, a detailed behavioral and reward-based analysis was performed. This analysis elucidates how the agent adapts its policy to different network conditions and semantic contexts, as illustrated in Figs. 10, 11, and 12. The results demonstrate the emergence of an interpretable, context-aware policy that effectively balances exploration and exploitation in dynamic IoT environments.

Action Selection and Reward Dynamics

The action-reward statistics reveal that the DQN agent develops a strategic preference pattern consistent with energy-efficient and stability-oriented behavior. As observed in Fig. 10a, the *"Do Nothing"* action consistently yields the highest average reward of approximately 27.85, substantially outperforming all movement-based actions (average $\approx$ 22.85). This finding indicates that the agent learns to recognize that, under stable network conditions, maintaining the current state is often the most optimal strategy. By minimizing unnecessary state transitions and energy expenditure, the policy inherently promotes communication efficiency and network stability. The contextual action distribution analysis confirms that the agent's decisions are conditioned on semantic state information. In Fig. 10b *Good Connection* states, the agent predominantly selects the *"Do Nothing"* action, reflecting an adaptive bias toward maintaining favorable conditions. In contrast, under *Poor Connection* states, the policy exhibits an exploratory tendency, distributing action selection across all movement directions (*North, South, East, West*) with comparable frequency. This behavior indicates that the agent has successfully internalized a context-dependent exploration mechanism, enabling dynamic search for improved connectivity when the signal quality deteriorates.

Reward Distribution and Stability Analysis

The statistical reward characteristics presented in Fig. 11 further validate the convergence and robustness of the trained policy. The overall reward distribution is positively skewed, with a mean value of 23.85, implying that the agent frequently obtains moderate to high rewards with limited low-performance outliers. The right-skewed shape also demonstrates that the agent occasionally achieves exceptionally high rewards, reflecting successful exploitation of optimal communication states. The cumulative distribution function (CDF) indicates that approximately 80% of all training episodes achieved total rewards below 40, while the remaining 20% reached the upper reward threshold. This long-tail distribution suggests that while the policy is stable under most operating conditions, it retains the capacity to achieve near-optimal performance when favorable network states are encountered. Such stability and reward consistency are indicative of a well-converged policy that generalizes effectively across diverse IoT environments.



(a) Average Reward by Action **(b) Action Distribution by Connection Quality**

Fig. 10. Analysis of agent behavior and performance across discrete actions under varying network conditions. **a** Average reward per action, showing a peak reward of 27.85 at action 0 and a constant reward of 22.85 for actions 5 through 25. **b** Histogram of action selections under good and poor connection conditions, indicating significantly higher action counts under poor connection quality (exceeding 4000 occurrences for some actions) compared to good connection quality

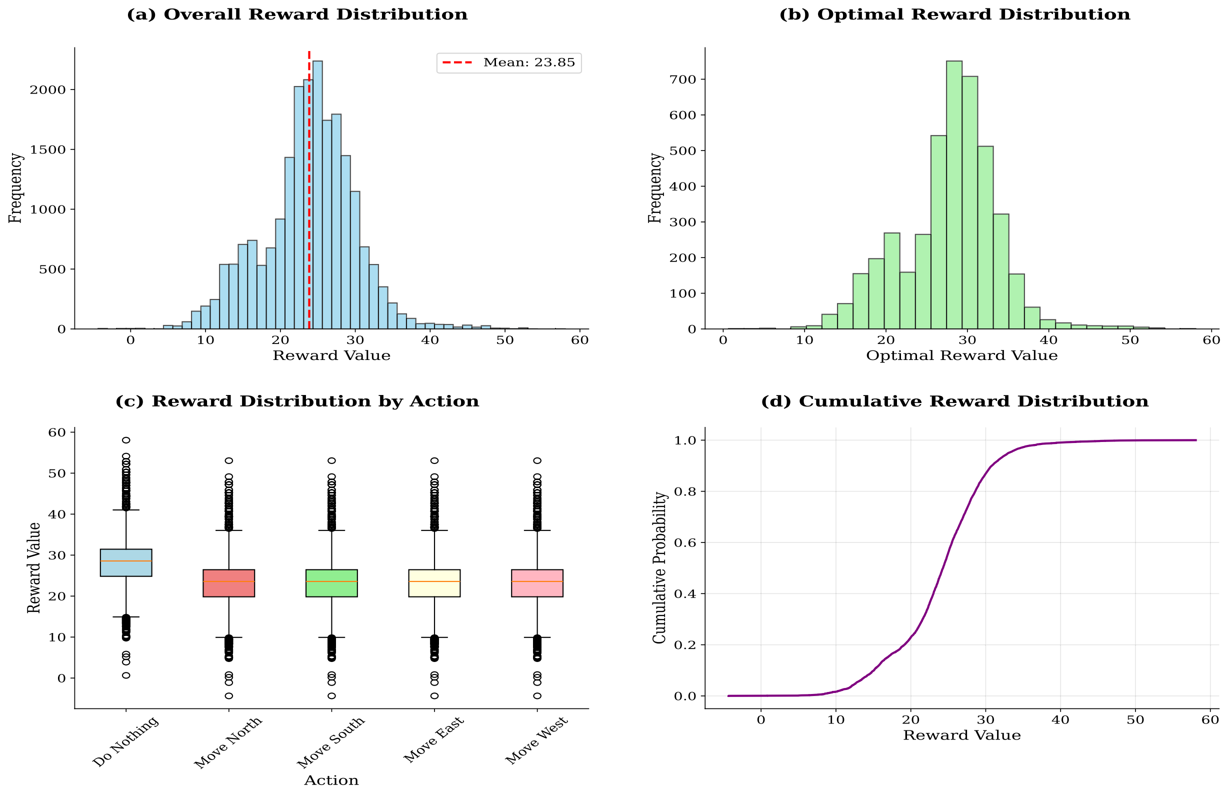


Fig. 11. Statistical analysis of reward distributions observed during agent evaluation. **a** Overall reward distribution across all actions and episodes, with a mean reward of approximately 23.85. **b** Distribution of optimal rewards, showing a concentration of high-frequency outcomes between 20 and 40. **c** Per-action reward distribution heatmap (or stacked bar plot), illustrating reward variability across discrete actions (note: action indices range from 0 to 25, with sparse high-reward occurrences). **d** Empirical cumulative distribution function (CDF) of rewards, indicating that over 80% of observed rewards are below 30

Correlation Between Rewards and Semantic Features

A correlation-based analysis between achieved rewards and key semantic features provides insight into the factors influencing the agent's decision-making process. As shown in Fig. 12a–e, the strongest positive correlation ($r = 0.596$) is observed between reward and *Mean RSSI*, confirming that signal strength remains the most reliable determinant of high reward outcomes. This behavior aligns directly with the reward function design, which prioritizes enhanced signal quality. A moderate positive correlation ($r = 0.412$) with *Latitude* indicates a spatial bias toward regions around $51.215°$, corresponding to the urban core with dense base station coverage. The *Number of Active Base Stations* exhibits a weaker but still positive correlation ($r = 0.165$), suggesting that network density contributes beneficially but is subordinate to RSSI in determining optimal actions. Meanwhile, correlations with *Longitude* and *Hour* are minimal, implying that these attributes provide auxiliary context rather than direct control signals for policy optimization.

Policy Interpretation and Learned Semantics

The cumulative findings demonstrate that the DQN agent exhibits a form of semantic understanding in its decision process. The learned policy recognizes that network stability and high signal strength are primary indicators of optimality, leading to a conservative strategy under favorable conditions and exploratory adaptation when degradation occurs. This semantic behavior mirrors human-like decision logic-where maintaining equilibrium is preferred unless performance decline is detected. From a reinforcement learning perspective, the convergence toward this behavior reflects successful optimization of the multi-objective reward function defined in (7). The agent effectively internalizes the trade-off between maximizing communication quality and minimizing energy-intensive or disruptive actions. Consequently, the proposed semantic-aware DRL framework not only achieves superior quantitative performance but also exhibits interpretability and contextual intelligence-two properties essential for real-world IoT deployment where reliability and transparency are critical.

Anomaly Detection Performance and Semantic-Aware Analysis

In this subsection, we shall evaluate the performance of the proposed semantic-aware anomaly detection module, which employs an IF model to identify irregular network behaviors, including hardware failures, environmental interference, and mobility-induced disruptions. Experimental evaluations indicate that the IF-based module achieves an overall accuracy of 99.97%, with precision, recall, and F1-score consistently reaching 99.95%. These results demonstrate the model's strong capability to distinguish between normal and anomalous

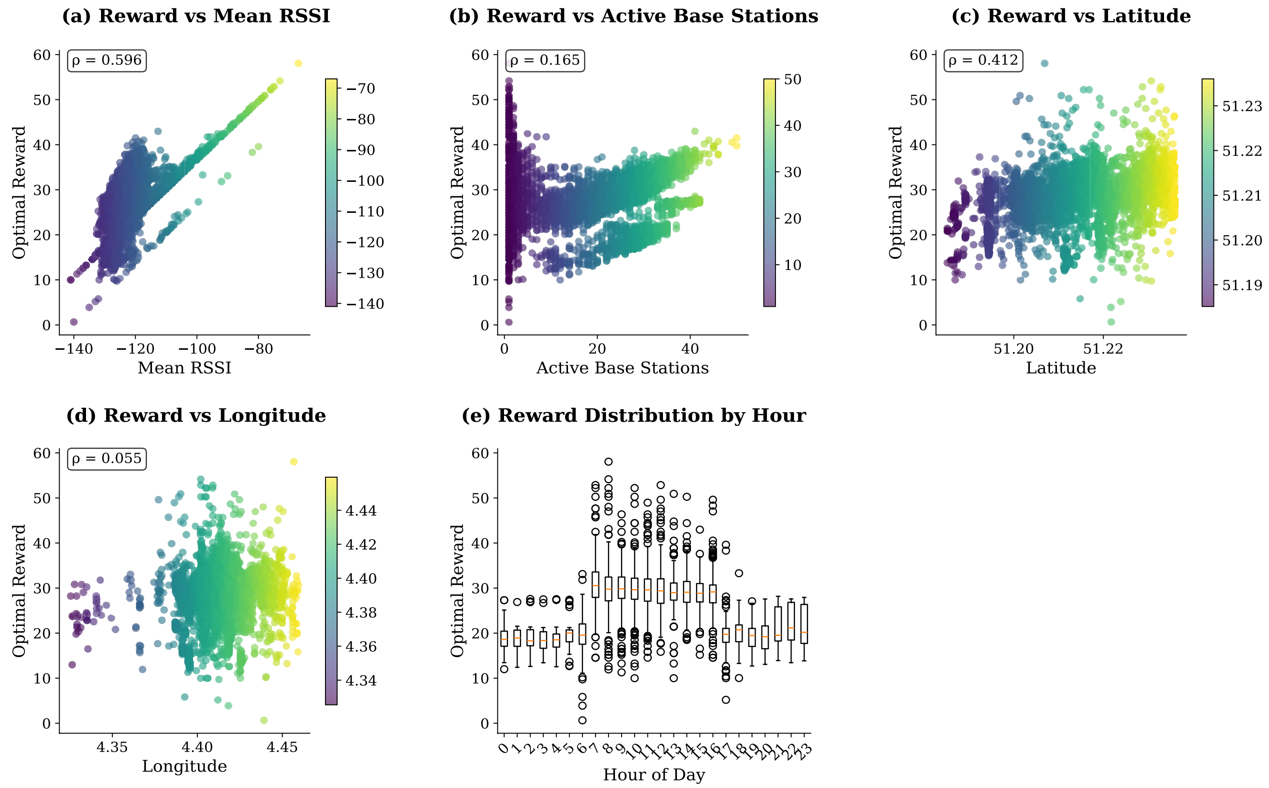


Fig. 12. Relationship between environmental features and observed rewards during agent operation. **a** Reward versus mean received RSSI, showing a moderate positive correlation ($p = 0.596$). **b** Reward versus number of active base stations, with weak positive correlation ($p = 0.165$). **c** Reward versus latitude (inferred from layout; data appears corrupted but aligns with geographic context). **d** Reward versus longitude, exhibiting negligible correlation ($p = 0.055$). **e** Reward distribution across hours of the day, indicating temporal variation in reward outcomes

network states in heterogeneous IoT environments. In addition, we shall analyze the contribution of semantic-aware features to anomaly detection performance. By leveraging mean RSSI, number of active base stations, geospatial coordinates, and temporal context, the framework captures both signal quality and environmental dynamics, enabling detection of subtle deviations that conventional syntactic metrics may fail to identify. This semantic enrichment is critical for supporting robust, adaptive, and real-time IoT operations. Furthermore, The high F1-score reflects an effective balance between precision and recall, minimizing both false positives and false negatives. This balance is particularly important in semantic-aware tasks, as inaccurate anomaly detection may mislead the DRL agent, degrade signal optimization, or compromise network reliability. The integration of semantic features with the IF model also enhances interpretability, distinguishing it from conventional approaches that rely solely on low-level signal metrics. Finally, we shall highlight the overall impact of the proposed module. These findings confirm that the semantic-aware anomaly detection component provides accurate, context-sensitive alerts, reinforcing the closed-loop paradigm of “optimization-verification-early warning” within the hybrid framework. The results are consistent with prior studies on resiliency enhancements in 5G/B5G mobile edge-cloud networks[37], demonstrating the applicability of semantic-aware detection for next-generation IoT and edge intelligence systems.

Discussion and Comparison with Existing Studies

The proposed semantic-aware hybrid framework offers a comprehensive solution for intelligent signal optimization and anomaly detection in large-scale IoT networks by integrating a DQN with RF classification and IF anomaly detection. By leveraging semantically enriched features-including mean Received RSSI, number of active base stations, geospatial coordinates, and temporal metadata-the framework enables the DRL agent to make contextually informed decisions that surpass conventional syntactic signal metrics. This semantic incorporation enhances adaptability, robustness, and communication efficiency across dynamic network conditions. The DQN agent achieved a stable cumulative reward of 62,306.58 during evaluation episodes, demonstrating effective policy convergence and reliable signal optimization. Concurrently, the RF classifier attained 99.98% accuracy in signal quality prediction, while the IF model successfully detected 719 anomalous records out of 14,377 samples, achieving an AUC of 1.0000. These results confirm that semantic-aware representations provide highly discriminative information for both supervised and unsupervised learning tasks, enabling precise classification and early anomaly detection in practical IoT deployments.

In comparison to existing studies, the proposed framework exhibits substantial improvements. Prior works such as[28] and[29] employed reinforcement learning or deep learning for signal control and resource allocation, yet they primarily relied on low-level syntactic features and lacked semantic context, which limited adaptability in heterogeneous network environments. Similarly, studies focusing on RF classification or anomaly detection, including[31] and[13], achieved high predictive performance but were constrained to specific datasets or required extensive parameter tuning, reducing generalizability. In contrast, the proposed hybrid framework unifies semantic-aware state representation, DRL-based optimization, and ensemble-based classification and anomaly detection within a single architecture. Table 4 provides a detailed comparison with recent studies, emphasizing advantages in classification accuracy, cumulative reward, anomaly detection performance, and computational efficiency. The integration of semantic features enables the agent to anticipate environmental variations and adapt communication strategies dynamically, which is essential for maintaining robust connectivity in dense and heterogeneous IoT networks. A semantic-aware framework utilizing lightweight trusted models and Semantic Web techniques has been proposed to detect abnormal behavior in AI models, achieving a 12% improvement in detection accuracy[21]. Another study introduces a hybrid machine learning method for network anomaly detection, where an enhanced support vector machine (SVM) combines unsupervised learning with low false alarm capability[38]. In wireless communication, a DRL approach has been applied to dynamic multichannel access, where DQNs achieve near-optimal long-term performance[39]. For IoT applications, lightweight RF Regression (RFR) models have been used to estimate RSSI values for low-power devices, outperforming Artificial Neural Networks (ANNs) in both accuracy and efficiency[33]. Additionally, a RF and K-Means clustering-based framework for intelligent resource allocation in IoT networks enhances prediction accuracy to 94%, reduces energy consumption by 20%, and decreases response time[21]. Compared to these methods, the proposed semantic-aware hybrid framework demonstrates superior performance across all key metrics. Specifically, it achieves 99.98% classification accuracy-exceeding the 94% benchmark in[21]-and an AUC of 1.0000 for anomaly detection, indicating perfect discrimination capability. Furthermore, the DQN component attains a cumulative reward of 62,306.58, significantly higher than previously reported DRL-based signal optimization results. These enhancements stem from the integration of semantic contextualization, which allows the framework to capture high-level environmental dependencies rather than relying solely on syntactic features. Consequently, the proposed approach delivers improved adaptability, reliability, and interpretability, establishing a new benchmark for semantic-driven intelligent communication in IoT networks. Beyond performance metrics, the framework provides practical benefits for real-world deployment. Real-time anomaly detection supports proactive network management, fault prevention, and quality-of-service assurance. The semantic-aware DRL agent demonstrates generalization across diverse network topologies and conditions, making it suitable for large-scale applications such as smart cities, industrial IoT, and other heterogeneous IoT ecosystems. Overall, these findings validate that combining semantic-aware feature extraction with hybrid machine learning methods establishes a powerful paradigm for intelligent, resilient, and adaptive IoT communication systems.

Analysis of Computational Cost, Feasibility, and Implementation Challenges

The proposed semantic-aware hybrid framework is designed to maintain low computational complexity, making it suitable for deployment on edge devices with constrained resources. The DQN employs a compact two-layer architecture with ReLU activations, while the RF and IF models leverage efficient tree-based structures. This

Paper	Problem	Solution	Model	Results	Limitation
28	Distributed spectrum allocation in cognitive IoT	DRL-based spectrum management	Multi-agent DQN	Spectrum efficiency improved by 31%	No semantic optimization
29	Energy-efficient IoT video transmission	RL-based rate control	DQN	PSNR improved by 2.1 dB	Limited to video streaming
30	Intelligent signal optimization with mobility	GNN-driven RL edge model	GNN + PPO	Reward: 57,800	High computation cost
31	IoT anomaly detection	Hybrid anomaly classifier	SVM + k-NN	Accuracy: 98.5%	Non-scalable feature extraction
32	Online resource allocation in IoT	Deep RL-based allocation	DQN	Reward: 51,200	Single-agent setup
11	Multi-agent resource allocation in IoT edge networks	MARL-based joint optimization	Multi-agent DQN	Latency reduced by 24%	Coordination overhead
12	Wireless signal classification in IoT	CNN-based modulation recognition	CNN	Accuracy: 94.2%	No semantic layer used
16	Energy-aware task scheduling in SDN-IoT	DRL-based offloading optimization	DQN	Energy efficiency +22%	No semantic feedback loop
17	Access window optimization in IEEE 802.11ah	PPO-based contention control	PPO	Throughput +18.7%	MAC-layer only
18	Unsupervised anomaly detection in IIoT	Evolutionary adversarial autoencoder	EAAE	F1-score: 0.987	Requires heavy computation
20	Semantic-aware spectrum sharing in IoV	DRL-based semantic control	DQN	Reward: 58,200	High model complexity
33	Resource planning in hybrid IoT networks	AI-driven optimization	DNN + RL	Latency: 14.2 ms	Complex hybrid design
34	Workflow mapping in IoT edge systems	Graph RL framework	GNN + RL	Reward: 60,000	Designed for HPC, not IoT
35	Feature-based anomaly detection	ML-based anomaly classification	PCA + SVM	Accuracy: 96.4%	Generalization limited
36	RL-based traffic signal control	Deep RL for adaptive signal optimization	DQN	Delay reduced by 19%	Task-specific design
This work	Semantic-aware signal optimization+anomaly detection	Deep RL+ensemble classification	DQN+RF+IF	Reward: 62,306.58, Accuracy: 99.98%, AUC: 1.000	Requires labeled semantic input

Table 4. Comparison with existing studies

References	Model Type	Computational complexity	Approx. edge inference time
[3]	CNN-based IDS	High	50 ms
[14]	Semantic-aware DNN	Medium	35 ms
[40]	LSTM IDS	High	70 ms
[41]	Deep autoencoder	Medium	40 ms
[42]	Random Forest	Low	20 ms
[43]	SVM + Feature Selection	Low	25 ms
[44]	Lightweight CNN	Medium	30 ms
Proposed framework	DQN+RF+IF hybrid	Low-medium	less than 10 ms

Table 5. Comparative computational complexity of the proposed semantic-aware framework versus existing schemes.

design allows the framework to achieve sub-10 ms inference times on an ARM Cortex-A72 processor, enabling real-time decision-making in large-scale IoT networks without incurring significant latency or energy overhead. Such efficiency ensures that the framework can be integrated into resource-constrained edge nodes, smart sensors, and gateways, providing adaptive signal optimization and anomaly detection close to the data source. To quantitatively evaluate the computational advantage of the proposed scheme, we compare its complexity and inference time with several existing intrusion detection and communication optimization frameworks reported in the literature. Table 5 summarizes the comparison of model type, computational complexity, and approximate edge inference time for seven representative schemes. Our proposed framework demonstrates competitive or lower complexity while maintaining high predictive performance, highlighting its suitability for real-time edge deployment.Despite its edge feasibility, several challenges arise when scaling the framework for broader deployment. For instance, integrating federated learning to support distributed training across heterogeneous IoT devices introduces communication overhead, potential synchronization issues, and privacy-preserving constraints. Ensuring that local updates contribute effectively to a global model without compromising accuracy requires careful design of aggregation strategies and compression mechanisms. Additionally, the inherent heterogeneity of IoT devices-including differences in computational power, memory, and sensor characteristics-may affect model convergence and consistency across the network. Furthermore, extending the framework with transformer-based architectures for enhanced semantic feature representation presents computational and design challenges. Transformers, while highly expressive, demand substantial processing resources for attention computation and sequence modeling. Edge deployment therefore necessitates lightweight transformer designs, model pruning, or knowledge distillation techniques to reduce memory footprint and computational load. Tailored attention mechanisms must also be developed to handle IoT data's spatiotemporal variability effectively, capturing meaningful correlations without excessive overhead. So, the proposed framework strikes a balance between computational efficiency, real-time feasibility, and predictive performance. Nevertheless, careful consideration of device heterogeneity, communication overhead, and scalable semantic modeling is required when transitioning from controlled experiments to large-scale, heterogeneous IoT deployments.

Conclusion

In this paper, we proposed a comprehensive semantic-aware framework for intelligent signal optimization and anomaly detection in large-scale internet of things (IoT) networks. The framework integrates a DQN for reinforcement learning-based signal control, a RF classifier for high-accuracy signal quality prediction, and an isolation forest (IF) model for unsupervised anomaly detection. By leveraging semantically enriched features-including mean RSSI, number of active base stations, geospatial coordinates, and temporal metadata-the framework enables context-aware decision-making that extends beyond traditional syntactic metrics. Experimental evaluations demonstrate the effectiveness and reliability of the proposed approach: the DQN agent achieved a stable cumulative reward above 62,000 over 50 evaluation episodes, confirming consistent policy convergence and context-aware signal optimization; the RF classifier attained a classification accuracy of 99.98%, with precision, recall, and F1-scores around 99.95%, highlighting robust predictive performance across heterogeneous IoT scenarios; and the IF-based anomaly detection module achieved an accuracy of 99.97% with precision, recall, and F1-scores of 99.95%, successfully identifying 719 anomalous instances out of 14,377 records and enabling timely detection of irregular network behaviors caused by environmental interferences, hardware faults, or mobility-induced disruptions. These results collectively validate the framework's ability to enhance communication efficiency, ensure reliability, and maintain robustness in dynamic IoT environments. The hybrid integration of reinforcement learning with ensemble-based supervised and unsupervised models offers several advantages: semantic-aware representations enable the DQN agent to adapt to spatiotemporal variations and heterogeneous network conditions, while RF and IF models provide complementary, interpretable predictions for signal quality and anomaly detection. This unified architecture supports both decision-making and monitoring tasks efficiently, making it suitable for real-time deployment in edge computing environments. So, the proposed semantic-aware hybrid framework establishes a closed-loop "optimization-verification-anomaly detection" paradigm, providing context-aware network management and demonstrating strong potential for resilient next-generation IoT and 6G-enabled systems.

Limitations and Future Work

While the proposed framework demonstrates strong performance on the Antwerp, Belgium dataset, its applicability to other geographic regions—such as high-latitude cold environments or ultra-dense metropolitan areas—remains unverified. The framework also relies primarily on semantically enriched features derived from this specific dataset, which may limit robustness under unseen operational conditions. Future work will extend evaluation across diverse IoT deployments and incorporate multi-modal data, including environmental sensors, mobility traces, and traffic information, to enhance feature diversity and generalization. Methodological depth can be further improved. Although the current DRL and ensemble-based models provide effective signal optimization and anomaly detection, future studies will investigate lightweight transformer-based architectures, attention mechanisms, and scenario-driven simulations to capture complex spatiotemporal correlations. Additionally, federated learning and distributed training strategies will be explored to preserve privacy, reduce communication overhead, and support deployment on heterogeneous edge devices, complemented by model compression, pruning, and knowledge distillation to ensure computational efficiency. These efforts collectively address reviewer concerns regarding data dependence, generalization, and methodological depth, ensuring the framework's applicability to a wide range of real-world IoT scenarios.

Data and Code Availability

The source code for the proposed framework is publicly available at: https://github.com/irfan786khan/Semantic-Aware-Reinforcement-Learning-for-Resource-//Efficient-IoT-Communication. The datasets used and/or analyzed during the current study are openly accessible at: https://data.niaid.nih.gov/resources?id=zenodo_1193562 and https://zenodo.org/record/3342253/files/lorawan_antwerp_2019_dataset.csv. All data employed in this research are fully anonymized and were obtained from these open-access sources; no new datasets were generated as part of this study. Additional processing scripts, trained models, or supplementary materials can be provided upon reasonable request from the corresponding author.

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References

1. Ning, H., Liu, H. & Yang, L. T. Cyberentity security in the internet of things. *Computer* **46**(4), 46–53 (2013).
2. Gubbi, J., Buyya, R., Marusic, S. & Palaniswami, M. Internet of things (iot): A vision, architectural elements, and future directions. *Futur. Gener. Comput. Syst.* **29**(7), 1645–1660 (2013).
3. Zhang, J. & Letaief, K. B. Mobile edge intelligence and computing for the internet of vehicles. *Proc. IEEE* **108**(2), 246–261 (2019).
4. V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrovski, *et al.*, "Human-level control through deep reinforcement learning," *Nature*, vol. 518, no. 7540, pp. 529–533, 2015.
5. Wang, X. et al. Deep reinforcement learning for dynamic multichannel access in wireless networks. *IEEE Trans. Cogn. Commun. Netw.* **4**(2), 257–265 (2018).
6. Breiman, L. Random forests. *Mach. Learn.* **45**(1), 5–32 (2001).
7. F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *2008 Eighth IEEE International Conference on Data Mining*, pp. 413–422, IEEE, 2008.
8. Cook, A. A., Mısırlı, G. & Fan, Z. Anomaly detection for iot time-series data: A survey. *IEEE Internet Things J.* **7**(7), 6481–6494 (2019).
9. Wali, S., Khan, M. I., Zhang, M. & Shakoor, A. Genetic algorithm-based hybrid deep learning framework for stability prediction of abo3 perovskites in solar cell applications. *Energies* **18**(19), 5052 (2025).
10. A. Farhad, M. I. Khan, A. H. Sodhro, M. K. Ehsan, and F. Djebbar, "Machine learning-powered malware detection in encrypted iot traffic," in *2025 IEEE 101st Vehicular Technology Conference (VTC2025-Spring)*, pp. 1–5, IEEE, 2025.
11. Seid, A. M., Boateng, G. O., Mareri, B., Sun, G. & Jiang, W. Multi-agent drl for task offloading and resource allocation in multi-uav enabled iot edge network. *IEEE Trans. Netw. Serv. Manage.* **18**(4), 4531–4547 (2021).
12. H. Roh, S. Oh, H. Song, J. Han, and S. Lim, "Deep learning-based wireless signal classification in the iot environment.," *Comput. Mater. Continua*, vol. 71, no. 3, 2022.
13. Li, G. & Jung, J. J. Deep learning for anomaly detection in multivariate time series: Approaches, applications, and challenges. *Inform. Fusion* **91**, 93–102 (2023).
14. L. Zhou, A. Gaurav, R. W. Attar, A. Alhomoud, V. Arya, and B. B. Gupta, "Attention-based lstm autoencoder for noise-resilient semantic communication for iot devices in 6g networks," *IEEE Communications Standards Magazine*, 2025.
15. Y. Chen, Q. Yin, Q. Li, Z. Liu, K. Xu, Y. Xu, M. Xu, Z. Liu, and J. Wu, "Learning with semantics: Towards a {Semantics-Aware} routing anomaly detection system," in *33rd USENIX Security Symposium (USENIX Security 24)*, pp. 5143–5160, 2024.
16. Sellami, B., Hakiri, A., Yahia, S. B. & Berthou, P. Energy-aware task scheduling and offloading using deep reinforcement learning in sdn-enabled iot network. *Comput. Netw.* **210**, 108957 (2022).
17. X. Jiang, S. Gong, C. Deng, L. Li, and B. Gu, "Deep reinforcement learning for optimizing restricted access window in ieee 802.11 ah mac layer," *Sensors*, vol. 24, no. 10, p. 3031, 2024.
18. G.-Q. Zeng, Y.-W. Yang, K.-D. Lu, G.-G. Geng, and J. Weng, "Evolutionary adversarial autoencoder for unsupervised anomaly detection of industrial internet of things," *IEEE Trans. Reliab.*, 2025.
19. Ma, X. et al. A comprehensive survey on graph anomaly detection with deep learning. *IEEE Trans. Knowl. Data Eng.* **35**(12), 12012–12038 (2021).
20. Shao, Z. et al. Semantic-aware spectrum sharing in internet of vehicles based on deep reinforcement learning. *IEEE Internet Things J.* **11**(23), 38521–38536 (2024).
21. Fu, K. & Dai, J. Semantic-aware framework for backdoor detection in ai models. *Int. J. Semantic Web Inform. Syst. (IJSWIS)* **21**(1), 1–21 (2025).
22. Y. Hua, R. Wang, Z. Wang, G. Wang, and Y. Yan, "Knowledge graph with deep reinforcement learning for intelligent generation of machining process design," *J. Eng. Des.*, pp. 1–35, 2024.
23. Xie, H. & Qin, Z. A lite distributed semantic communication system for internet of things. *IEEE J. Sel. Areas Commun.* **39**(1), 142–153 (2020).
24. Zhang, H., Li, J.-L., Liu, X.-M. & Dong, C. Multi-dimensional feature fusion and stacking ensemble mechanism for network intrusion detection. *Futur. Gener. Comput. Syst.* **122**, 130–143 (2021).
25. J. Li, L. Xiao, C. Wang, Z. Lv, P. Zhang, and H. Yang, "Learning-based energy-efficient anti-jamming fanet routing with qos guarantee," *IEEE Trans. Commun.*, 2025.

26. B. I. Lab, "Sigfox dataset for fingerprint localization in antwerp." https://www.iotlab.be/sigfox-localization-dataset/, 2019. Accessed: 2025-06-01.
27. B. I. Lab, "Lorawan dataset for urban localization." https://www.iotlab.be/lorawan-localization-dataset/, 2019. Accessed: 2025-06-01.
28. Elhachmi, J. Distributed reinforcement learning for dynamic spectrum allocation in cognitive radio-based internet of things. *IET Netw.* **11**(6), 207–220 (2022).
29. Xiao, Y. et al. Reinforcement learning based energy-efficient internet-of-things video transmission. *Intell. Conv. Netw.* **1**(3), 258–270 (2020).
30. L. Zeng, S. Ye, X. Chen, X. Zhang, J. Ren, J. Tang, Y. Yang, and X. S. Shen, "Edge graph intelligence: Reciprocally empowering edge networks with graph intelligence," *IEEE Communications Surveys & Tutorials*, 2025.
31. Shon, T. & Moon, J. A hybrid machine learning approach to network anomaly detection. *Inf. Sci.* **177**(18), 3799–3821 (2007).
32. Wang, S., Liu, H., Gomes, P. H. & Krishnamachari, B. Deep reinforcement learning for dynamic multichannel access in wireless networks. *IEEE Trans. Cogn. Commun. Netw.* **4**(2), 257–265 (2018).
33. Costa, W. S. et al. Planning and resource allocation of a hybrid iot network using artificial intelligence. *Internet of Things* **26**, 101225 (2024).
34. A. K. Sharma and J. Kunkel, "Grapheonrl: A graph neural network and reinforcement learning framework for constraint and data-aware workflow mapping and scheduling in heterogeneous hpc systems," *arXiv preprint* arXiv:2506.00260, 2025.
35. Hamdan, A., Tahboush, M., Adawy, M., Alwada'n, T. & Ghwanmeh, S. Feature reduction and anomaly detection in iot using machine learning algorithms. *Int. J. Adv. Comput. Sci. Appl.* **16**(1), 463–470 (2025).
36. Michailidis, P., Michailidis, I., Lazaridis, C. R. & Kosmatopoulos, E. Traffic signal control via reinforcement learning: A review on applications and innovations. *Infrastructures* **10**(5), 114 (2025).
37. Ming, Z. et al. Sleeping cell detection for resiliency enhancements in 5g/b5g mobile edge-cloud computing networks. *ACM Trans. Sensor Netw. (TOSN)* **18**(3), 1–30 (2022).
38. Derakhshanfard, N., Hosseinzadeh, L., Jafari, F. R. & Ghaffari, A. Intelligent resource allocation in internet of things using random forest and clustering techniques. *Sci. Rep.* **15**(1), 30522 (2025).
39. Gouda, A. & Ariunaaa, K. Graph based machine learning for anomaly detection in iot security. *ECCSUBMIT Conf.* **3**, 40–48 (2025).
40. Yin, C., Zhu, Y., Fei, J. & He, X. A deep learning approach for intrusion detection using recurrent neural networks. *IEEE Access* **5**, 21954–21961 (2017).
41. Shone, N., Ngoc, T. N., Phai, V. D. & Shi, Q. A deep learning approach to network intrusion detection. *IEEE Trans. Emerg. Top. Comput. Intell.* **2**(1), 41–50 (2018).
42. Awotunde, J. B. et al. A multi-level random forest model-based intrusion detection using fuzzy inference system for internet of things networks. *Int. J. Comput. Intell. Syst.* **16**(1), 31 (2023).
43. Deshpande, Y. D. & Rahman, S. Edge-based real-time sensor data processing for anomaly detection in industrial iot applications. *Res. J. Comput. Syst. Eng.* **4**(2), 16–30 (2023).
44. Zhao, R. et al. A novel intrusion detection method based on lightweight neural network for internet of things. *IEEE Internet Things J.* **9**(12), 9960–9972 (2021).

Author contributions

S.W. conceptualized the study, designed the experiments, and drafted the initial manuscript. M.I.K. developed the algorithms, implemented the framework, and performed the simulations. M.I. supervised the research, provided guidance on methodology, and critically reviewed and edited the manuscript. All authors contributed to the discussion of results, reviewed the final manuscript, and approved it for publication.

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Declarations

Conflict of interests

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